Monday, January 16, 2023 5:12 PM

Selenium Webdriver – Introduction

Selenium Webdriver is a robust tool for testing the front end of an application and to perform tasks on the browser. Selenium tests can be created in multiple programming languages like Python, Java, C#, JavaScript, and so on. Selenium with Python combination is comparatively easy to understand and it is short in verbose.

The APIs available in Python enable us to create a connection with the browser using Selenium. Selenium provides various Python commands which can be used for creating tests for different browsers like Chrome, Firefox, IE, and so on. It can be used in various platforms like Windows, Mac, Linux, and so on.

Reasons to learn Selenium with Python

 Python is easier to learn and compact in terms of programming.

 While creating tests in Selenium with Java, we have to take care of the beginning and ending braces. In Python, simply code indentation needs to be taken care of.

 Tests developed in Selenium with Python run faster than those written in Java. Reasons to learn Selenium Webdriver The reasons to learn Selenium Webdriver are mentioned below: It is open source and comes without any licensing cost.

 It can perform mouse and keyboard actions like drag and drop, keypress, click and hold, and so on. It has a very friendly API.

 It can be integrated with frameworks like TestNG and JUnit, build tools like Maven, continuous integration tools like Jenkins.

 It has a huge community support. It can execute test cases in headless mode

Selenium Webdriver — Browser Navigation

We can open a browser and navigate to an application with the help of Selenium webdriver in Python. This is done with the help of the get method. While automating a test, the very first step that we create is launching an application with a URL.

Syntax:-
driver.get("")

For a get method, the webdriver waits till the page is completely loaded before moving to the next step. If we try to launch a web page having numerous AJAX calls, then the webdriver is unaware when the page is completely loaded. To fix this issue, we have to apply waits in our code.

```
fromseleniumimportwebdriver
fromselenium.webdriver.chrome.serviceimportService
fromselenium.webdriver.common.byimportBy
importtime

service_obj=Service("C:\\ChromeDriver\\chromedriver.exe")
driver=webdriver.Chrome(service=service_obj)

driver.get('https://www.facebook.com/r.php?locale=en_GB&display=page')

print(driver.title)

time.sleep(100)
driver.close()
```

Output

The output is given below:

```
/Users/debomitabhattacharjee/PycharmProjects/pythonProjectTest/venv/bin/python
Page title: The Best Technical Questions And Answers

Process finished with exit code 0
```